



Guided contrastive boundary learning for semantic segmentation

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ABSTRACT

Semantic segmentation, a fundamental task in environmental understanding, aims to assign each image pixel to a specific class. Despite recent progress, segmentation accuracy in boundary regions remains suboptimal. This paper introduces Guided Contrastive Boundary Learning (GCBL), a novel framework designed to enhance feature representation learning, thereby improving boundary segmentation performance. Unlike conventional contrastive learning, GCBL guides inter-class representation learning by weighting pixel contributions based on their estimated probabilities. For intra-class learning, it leverages the neural collapse phenomenon, encouraging representations to align with last-layer classifier weights. Additionally, an asymmetric distance boundary pixel search strategy ensures a more reasonable selection of contrastive pairs. To prevent weight collapse in learning, a regularization term is applied to the last-layer classifier's weights. The GCBL method is readily integrable into existing and future segmentation frameworks. Extensive experiments on the Cityscapes, ADE20K, and S3DIS datasets demonstrate the effectiveness and generalizability of our approach. Code is available at <https://github.com/skyshoumeng/GCBL>.

1. Introduction

Semantic segmentation is a fundamental yet challenging task with applications in diverse domains such as autonomous driving, video surveillance, and medical image analysis [1–4]. Its objective is to assign each pixel in an image to a predetermined class. Despite significant advancements in the field, the performance of segmentation methods remains suboptimal in boundary regions [5,6]. However, accurate boundary segmentation is crucial in real-world scenarios like autonomous driving. For instance, erroneous boundary segmentation can severely impair the recognition of small objects with limited pixel or point representations, such as pedestrians, cyclists, and traffic sign poles. This can ultimately result in collisions with curbs and other road traffic accidents.

Segmentation at boundary areas presents a greater challenge than segmentation within object interiors. Typically, feature representations at the output layer are often derived from multiple convolutional layers and spatial upsampling interpolation operations [7,8]. Consequently, the feature representation of each pixel incorporates substantial neighborhood or contextual information. While this context aids in segmenting inner regions [7], it leads to smoothed feature representations at boundaries, diminishing their discriminability [9].

In general, current methods lack an explicit and comprehensive investigation of feature representation learning at boundary areas [7,9]. For instance, in STDC [7], the detail guidance module aims to guide low-level layers to learn spatial information, such as boundaries and corners. However, this module is simply supervised by the boundary labels derived from segmentation labels using the Laplacian operator, without further exploration. Similarly, boundary-aware segmentation methods like [10] impose boundary distance map supervision on the final predictions, but do not explicitly investigate feature representation learning at the boundaries. While CBL [9] explicitly considers feature learning at boundaries, the direct application of contrastive learning to segmentation tasks poses challenges. One challenge is that feature representations of neighboring pixels from different classes are pushed apart with equal force, even though some representations may already be adequate and should not be treated equally. Additionally, contrastive learning is performed only at the local boundary area, lacking global information guidance.

Therefore, to address the challenges of applying contrastive learning to segmentation tasks, we propose a novel Guided Contrastive Boundary Learning (GCBL) approach that significantly improves semantic segmentation performance at boundary regions. As illustrated in Fig. 1, our method incorporates guidance information into the contrastive

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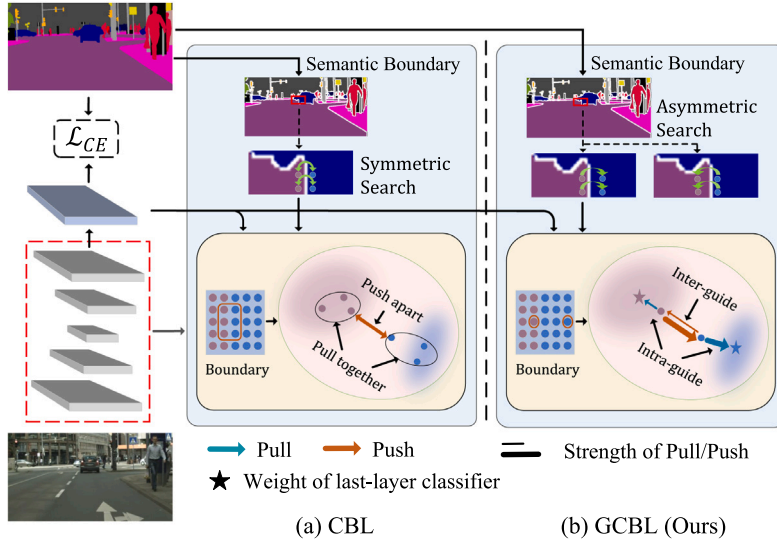


Fig. 1. Our proposed GCBL framework adopts an asymmetric boundary pixel search strategy to facilitate more judicious pixel selection. By incorporating guidance information into contrastive boundary learning for both inter-class and intra-class learning, GCBL effectively enhances feature representation learning and refines boundary segmentation.

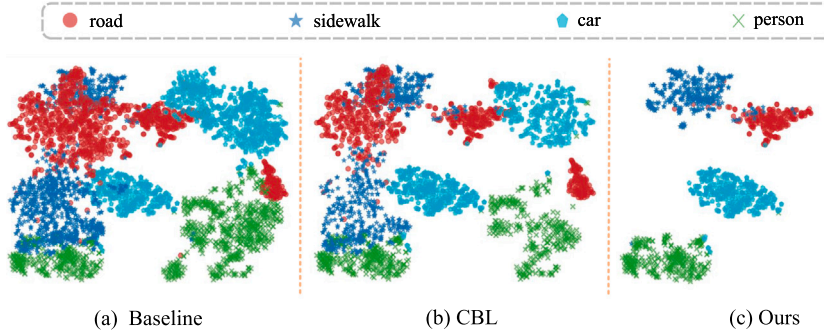


Fig. 2. Visualizations of logit embeddings at the boundary areas under different training losses. It can be seen that our proposed Guided Contrastive Boundary Learning (GCBL) approach can enhance feature representation learning, achieving better separation of feature embeddings between different classes. For fairness of comparison, the experiments are conducted on all the images of the Cityscapes validation sets.

learning process. For inter-class representation learning, we weight the contribution of each pixel in a contrastive pair by the estimated probability of the model belonging to its ground-truth semantic class, referred to as the strength of push (lower right of Fig. 1(b)). For intra-class representation learning, we utilize the weights in the last-layer classifier as guidance, leveraging the phenomenon of neural collapse [11].

Additionally, we introduce an asymmetric boundary pixel search algorithm for a more reasonable selection of contrastive pairs. We experimentally observed that our contrastive learning loss could potentially lead to a collapse of the weight distribution in the last-layer classifier [11], resulting in less diverse outputs and reduced model robustness. To mitigate this issue, we propose a regularization loss that encourages a more distributed weight distribution in the last-layer classifier. Fig. 2 presents a visual comparison of feature representation learning results among the baseline method, the CBL method, and our proposed method. It is evident that our GCBL achieves a superior feature distribution across different classes.

Finally, to evaluate the effectiveness and generalization of our proposed method, we conducted experiments with six baselines across three challenging datasets. These include image-based baselines STDC [7], SERT [12], SegFormer [8], DeepLabV3 [13], OCRNet [14], and the point-based baseline CBL [9]. The datasets used for evaluation were Cityscapes [15], ADE20K [16], and S3DIS [17]. The experimental results consistently demonstrate performance improvement across all evaluated datasets.

In summary, our contributions include the following:

1. We identify and analyze the challenges associated with boundary segmentation in current methods and propose a novel guided contrastive boundary learning (GCBL) approach. This approach effectively enhances feature representation learning and segmentation performance at boundary areas. The improved feature representation learning is visually confirmed through t-SNE visualizations, demonstrating a clearer separation of classes at boundaries.
2. We introduce an asymmetric distance boundary pixel search strategy, enabling a more rational selection of contrastive pairs and further enhancing the effectiveness of feature representation learning.
3. To address the issue of weight collapse in the last-layer classifier during contrastive learning, we propose a regularization loss that encourages a more distributed weight distribution, improving model robustness.
4. We conduct extensive experiments on six state-of-the-art deep learning-based semantic segmentation architectures across diverse datasets. The results consistently demonstrate significant performance improvements when our proposed GCBL approach is incorporated.

2. Related work

In this section, we briefly introduce the related works of our proposed approach: Boundary-aware Segmentation and Contrastive Learning.

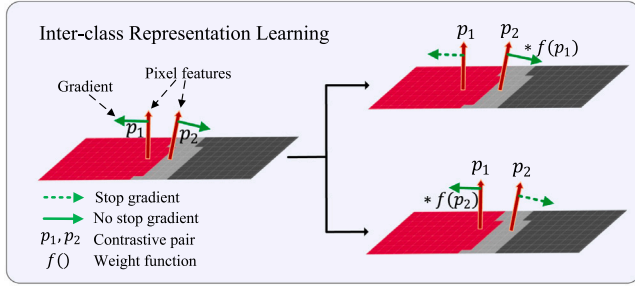


Fig. 3. Our guided inter-class representation learning scheme. Unlike conventional contrastive learning, which applies equal force to push apart features from pixels of different classes, our approach divides learning into two asymmetric parts. In each part, one side of the boundary (reference pixels) has fixed gradients, while the push strength on the other side (learning pixels) is weighted by the model's estimated probability of the reference pixel's class.

2.1. Boundary-aware segmentation

In recent years, there has been growing research interest in boundary-aware segmentation. These methods can be broadly categorized into two categories: boundary refinement methods and boundary-based loss methods.

Boundary refinement methods have garnered significant attention due to their demonstrated effectiveness in improving segmentation accuracy at object boundaries. BMask R-CNN [5] incorporates an additional branch for direct boundary estimation, thereby enhancing feature representations in these regions. SegFix [18] proposes replacing unreliable boundary pixel predictions with more reliable interior pixel predictions. BPR [19] introduces a post-processing scheme that refines predicted instance boundaries to improve segmentation mask quality. RefineMask [20] integrates fine-grained features during a multi-stage instance-wise segmentation process, consistently producing refined, high-quality masks.

Boundary-based loss functions, such as the distance map penalized cross-entropy loss [10] and the Hausdorff distance loss [21], have emerged as a promising approach for improving segmentation accuracy at object boundaries. These methods aim to minimize the discrepancy between ground truth and predicted segmentations in boundary regions, with a particular emphasis on hard-to-segment areas. ABL [22] further advances this approach by progressively aligning predicted and ground-truth boundaries during end-to-end training.

While these methods have demonstrated notable performance gains, they generally lack an explicit focus on the underlying feature representation learning at boundary areas. This leaves a gap in understanding how the learned features contribute to improved boundary segmentation, potentially limiting further advancements.

2.2. Contrastive learning

Contrastive learning methods have shown promising results across a range of unsupervised and supervised computer vision tasks [23, 24], often employed as a pre-training step to enhance image embedding [25,26]. By comparing similar (positive) and dissimilar (negative) data pairs, these methods learn discriminative representations. Recently, contrastive learning has been extended to both 2D image segmentation [26,27] and 3D point cloud segmentation [28,29].

For example, in [25], a novel contrasting strategy leverages structural similarity and a localized contrastive loss to learn distinctive representations of local regions. DenseCL [26] introduces a dense self-supervised learning method, optimizing a pairwise contrastive (dis)similarity loss between two views of input images at the pixel level. In [27], a fully supervised pixel-wise contrastive learning algorithm is proposed for semantic segmentation, explicitly utilizing the

structure of labeled pixels. Recently, CBL [9] introduces a contrastive boundary learning framework for point cloud semantic segmentation that effectively improves feature discrimination across boundaries by leveraging multi-scale scene contexts. However, this approach has potential limitations. In inter-class learning, the uniform push-apart operation on pixel feature pairs may inadvertently disrupt previously well-learned representations, as it does not differentiate between features needing adjustment and those already accurate. Additionally, while the intra-class pull-together operation enhances local feature consistency, it relies solely on local information, limiting its effectiveness for minimizing distance between features within the same semantic class.

3. Proposed method

In this section, we begin by presenting an overview of our guided contrastive boundary learning (GCBL) approach. We then delve into the details of the proposed guided inter-class representation learning and guided intra-class representation learning components. Next, we provide a comprehensive introduction to the asymmetric boundary pixel search strategy for contrastive-pair selection, a critical aspect of contrastive learning. Finally, we analyze the weight collapse problem observed in our contrastive learning process and propose a weight regularization loss to mitigate this issue.

3.1. Guided contrastive learning

The proposed guided contrastive boundary learning approach comprises two components: guided inter-class representation learning and guided intra-class representation learning.

3.1.1. Guided inter-class representation learning

For the inter-class representation learning, most previous contrastive learning methods [9,27] focus on increasing the distance between features from different classes. Given all pixel pairs $\{(p_1, p_1^n), (p_2, p_2^n), \dots, (p_N, p_N^n)\}$, the goal of contrastive inter-class learning is to increase the distance of learned representations between neighboring pixels from different classes. The loss function can be formulated as follows:

$$\mathcal{L}_{inter} = \frac{1}{|\mathbb{B}|} \sum_{i \in \mathbb{B}} \phi(M(p_i, p_i^n)), \quad (1)$$

where $\mathbb{B}$ is the semantic boundary area obtained from labels and $|\mathbb{B}|$ denotes the number of pixels in the boundary. $M(\cdot, \cdot)$ is a distance measurement function, and $\phi(\cdot)$ is a monotonically decreasing function.

However, previous approaches overlook the information inherent in pairs. Consider the example of pixels p_1 and p_2 located on opposite sides of the boundary in Fig. 3. Assuming p_1 is predicted correctly with high probability, and the feature of p_2 closely resembles that of p_1 due to spatial proximity, it implies the prediction of p_2 is far from correct. In this case, simply increasing the feature distance between p_1 and p_2 is inappropriate, as it could compromise the accurate representation of p_1 . Pushing p_1 away from p_2 based on the inaccurate representation of p_2 may lead to learning issues [30]. Our experiments also confirmed that this approach results in ineffective representation learning.

In response, we incorporate the prediction information of each pixel into our inter-class learning approach. Specifically, we divide inter-class learning into two parts. In each part, one side of the contrastive pairs is designated as the reference side, and the other side is taken as the learning side. The contribution of each part to the final contrastive learning loss is weighted by the model's estimated probability of the corresponding reference pixels belonging to the ground truth class. Higher estimated probabilities for the reference pixels result in a larger scaling coefficient for their contribution, and vice versa. The details of

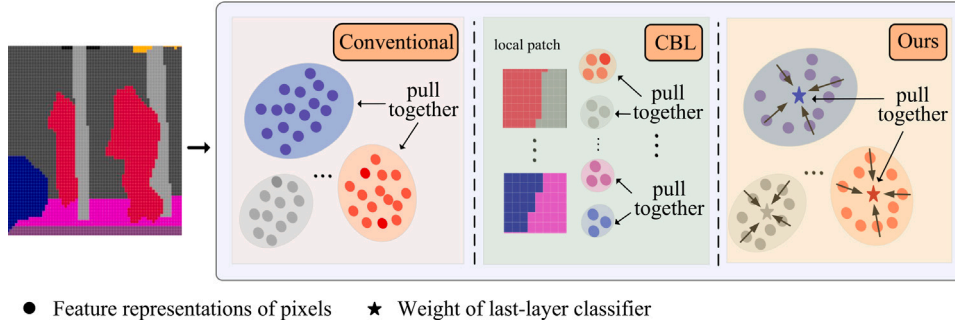


Fig. 4. Comparison between the intra-class representation learning process of conventional contrastive learning, contrastive boundary learning (CBL), and our proposed guided intra-class representation learning. We utilize the weight of last-layer classifier as the global guidance to boost intra-class representation learning rather than only use local features at boundary areas in the CBL method.

our guided inter-class representation learning are illustrated in Fig. 3, and the loss function is expressed as follows:

$$\mathcal{L}_{inter} = \frac{1}{|\mathbb{B}|} \sum_{i \in \mathbb{B}} \phi(M(p_i, D(p_i^n))) \cdot D(f(p_i^n)) + \frac{1}{|\mathbb{B}|} \sum_{i \in \mathbb{B}} \phi(M(D(p_i), p_i^n)) \cdot D(f(p_i)), \quad (2)$$

where $D(\cdot)$ denotes the detach from gradient operation, and $f(\cdot)$ denotes the function used to reweight the loss contribution. We denote $f(p_i)$ and $f(p_i^n)$ as the model's estimated probabilities of the reference pixels p_i and p_i^n , respectively.

From Eq. (2), we can observe that if the prediction at p_i is far from correct, p_i will be pushed away from p_i^n with a weak force. Conversely, if the prediction at p_i is correct with high probability, then p_i will be pushed away from p_i^n with a stronger force.

3.1.2. Guided intra-class representation learning

For intra-class representation learning, conventional contrastive learning methods aim to minimize the distance between samples from the same class. However, CBL [9] limits this encouragement to samples in nearby boundary areas. This approach is less effective because the feature representations of pixels from nearby areas are already highly similar due to spatial proximity. A potential solution is to identify all pixels belonging to each class in each batch and then encourage the learned representations of each class to become more similar through a carefully designed loss function. Unfortunately, this approach is computationally expensive due to the large number of contrastive pairs and the associated distance calculations.

To address the aforementioned issue, we observe that predictions of pixels far from boundaries are more likely to be correct, while those near boundaries are often incorrect. Therefore, we hypothesize that only pixels near the boundary are necessary for contrastive boundary learning. However, even in this scenario, the large number of pixels selected for intra-class learning still poses significant computational challenges in calculating pairwise distances. To overcome this, inspired by the theoretical analysis of deep learning training progress in [11], where last-layer classifier weights collapse to class means, we transform the task of reducing variance between classes into reducing the distance between pixel features of each class and their respective class means. In practice, these class means are replaced with the weights of the last-layer classifier, rendering the computational cost negligible. Fig. 4 illustrates the comparison between the CBL approach and our guided intra-class representation learning. For reference, the intra-class loss in the CBL framework is expressed as:

$$\mathcal{L}_{intra} = \frac{1}{|\mathbb{B}|} \sum_{i \in \mathbb{B}} \sum_{p_j \in \mathcal{N}_i \cap I_j = I_i} \phi(M(p_i, p_j)), \quad (3)$$

where $\mathcal{N}_i$ is the local neighborhood of point p_i , I_i is the label, and $\phi(\cdot)$ is a monotonically increasing function. In our method, the loss becomes:

$$\mathcal{L}_{intra} = \frac{1}{|\mathbb{B}|} \sum_{i \in \mathbb{B}} \phi(M(D(w_i), p_i)), \quad (4)$$

where w_i is the weight corresponding to the class of p_i in the last-layer classifier. Here, we employ the stop gradient operation on the weight parameters, as our objective is to enhance the feature representation of each pixel rather than adjusting the model parameters. Moreover, we experimentally found that this can enhance the stability of training.

For the distance measurement function $M(\cdot, \cdot)$, we simply adopt the commonly used cosine similarity function: $M(p_i, p_i^n) = \cos(p_i, p_i^n) = \frac{p_i \cdot p_i^n}{\|p_i\| \|p_i^n\|}$. However, the norm of the vector also has an impact on the distance between the features. Hence, we integrate both the angles and the norm distance to comprehensively consider the distance relationship between vectors. We aim to increase the angle between the negative pairs while keeping the norm of the feature vectors as the same as possible [31]. Specifically, we use cosine similarity as a measure of distance and take the norm part as a regularization term. The regularization term is formulated as follows:

$$\mathcal{L}_{reg} = -\frac{1}{|\mathbb{B}|} \sum_{i \in \mathbb{B}} \lambda \log \left(\min \left(\frac{\|p_i\|}{\|p_i^n\|}, \frac{\|p_i^n\|}{\|p_i\|} \right) \right), \quad (5)$$

where λ is the weight parameter to control the balance between the distance loss and regularization loss. Here, we use $\min(\cdot)$ operation to ensure that the regularization loss is always positive. In our experiments, we set $\phi(x) = 1 - x$ and $\varphi(x) = x$ for Eqs. (2) and (4).

3.2. Asymmetric distance search

Contrastive-pair selection is a crucial and challenging task in contrastive learning [32,33], and this also holds true for contrastive boundary learning. Previous works defined a pixel as a boundary pixel if a nearby pixel had a different class label, enabling easy identification of boundary pixels from the ground-truth label. Intuitively, two boundary pixel selection strategies exist. One strategy selects pixels very close to the boundary, as shown in Fig. 5(a). However, this approach has two issues. First, prediction errors often occur in these boundary-close areas, making it difficult to derive useful guidance from these pixels. Second, the high similarity of features between adjacent pixels causes the gradient corresponding to the Guided Inter-class Representation Learning loss $\mathcal{L}_{inter}$ to nearly vanish. The reasons for this are detailed below: for a given pair of pixels p_i and p_i^n , located on either side of the boundary and very close to each other, the gradient of the cosine similarity part of $\mathcal{L}_{inter}$ loss can be evaluated as follows:

$$\begin{aligned} \frac{\partial \mathcal{L}_{inter}}{\partial p_i} &= \frac{\partial (\phi(M(p_i, D(p_i^n))) \cdot D(f(p_i^n)))}{\partial p_i} \\ &\quad + \frac{\partial (\phi(M(D(p_i), p_i^n)) \cdot D(f(p_i)))}{\partial p_i} \\ &= \frac{1 - \cos(f_i, D(f_i^n))}{\partial p_i} \cdot D(f(p_i^n)) \\ &= \sin(f_i, D(f_i^n)) \cdot D(f(p_i^n)) \end{aligned} \quad (6)$$

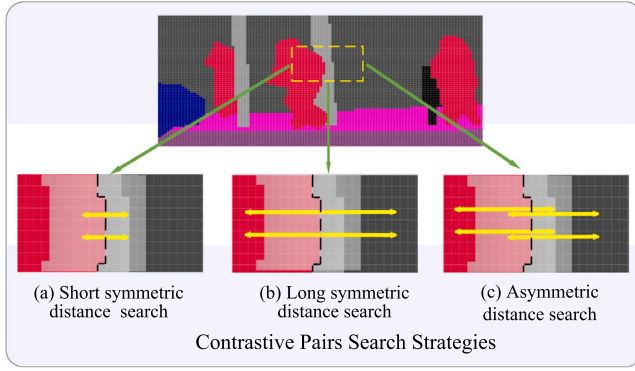


Fig. 5. Comparison with different search strategies. (a) Symmetric short distance search strategy; (b) Symmetric long distance search strategy; (c) Proposed asymmetric distance search. Two times asymmetric distance searches were conducted at the same boundary location, with one side (long distance) as the reference side (detached from the gradient) and the other side (short distance) as the learning side (no detachment) each time.

where $\sin(\cdot, \cdot)$ denotes the sine value of the angle between two vectors. We note that the gradient of the cosine similarity part of $\mathcal{L}_{inter}$ is proportional to $\sin(f_i, D(f_i^n))$. Therefore, when the cosine similarity between two neighboring features p_i and p_i^n approaches to 1, as is often the case, the gradient nears zero, making the training inefficient.

The other feasible approach is selecting pixels that are far away from the boundary, as shown in Fig. 5(b). However, this strategy is not suitable for fine and elongated objects, such as distant pedestrians and traffic sign poles, as it may fail to select their corresponding boundary pixels.

To address the limitations of the symmetric search method, we propose a simple yet effective boundary pixel selection scheme with asymmetric distances. As illustrated in Fig. 5(c), we sample twice at the same location within the boundary area. Each time, one side of the boundary pixels is designated as reference pixels (sampled at a long distance), while the other side is designated as learning pixels (sampled at a short distance), corresponding to the two parts described in Section 3.1.1. In this scenario, the reference pixels are more likely to yield accurate predictions, providing valuable information for contrastive learning. Conversely, the features of learning pixels are often suboptimal and require updating.

In our experiments, we select pixels in the direction of the boundary normal as contrastive pairs. To achieve this, we first perform boundary detection on the ground truth label, followed by directional filtering to obtain the normal direction of the boundary. This filtering is efficiently implemented using standard convolution operations. After obtaining the normal vector, we conduct an asymmetric search operation on both sides of the boundary. Note that the asymmetric search is employed for inter-class representation learning, while for intra-class learning, we only consider the reference pixels without gradient detachment. The complete search procedure is summarized in Algorithm 1.

3.3. Classifier weight regularization

During training with the aforementioned supervision, we observed a phenomenon contrary to [11]. Specifically, instead of the last-layer classifier weights collapsing towards the vertices of a Simplex Equiangular Tight Frame (ETF), we found that the cosine similarity between these weights gradually increased. This resulted in poor model generalization. We explain this as follows: when weights have high cosine similarity, prediction results are highly sensitive to small changes in feature representation due to the reduced feature space between

Algorithm 1 The Proposed Asymmetric Distance Negative Pairs Search

Require: Image I ; Gaussian smooth filter S ; Directional filters $\{F_x, F_y\}$; Ground-truth label G ; Asymmetric distance $\{d_1, d_2\}$;

Ensure: Boundary pixel pairs;

- 1: Detect the boundary on the Ground-truth label G to obtain B ;
- 2: Smooth the boundary map B with the Gaussian smooth filter S ;
- 3: Generate the normal vector of the boundary by performing the convolution operation with the directional filters.
- 4: Search along both sides of the normal vector with the predefined asymmetric distance: d_1, d_2 , obtain boundary pixel pairs $\{P, P_n\}$;
- 5: Return P, P_n .

classes. Conversely, when weights have lower cosine similarity, predictions are more robust to changes in feature representation, as the inter-class feature space is larger. We attribute this issue to the guided intra-class learning loss, as larger cosine similarity among weights implies a smaller feature space for each class, and consequently, smaller distances between pixels of the same class. To counteract this problem, we introduce a regularization loss that encourages greater distance between weights. Considering both the angle and norm of the weights, this regularization loss is formulated as:

$$\mathcal{L}_W = \frac{1}{|\mathcal{C}|^2} \sum_{i \in \mathcal{C}} \sum_{j \in \mathcal{C}} \cos(w_i, w_j) - \lambda \log \left(\min \left(\frac{\|w_i\|}{\|w_j\|}, \frac{\|w_j\|}{\|w_i\|} \right) \right), \quad (7)$$

where $|\mathcal{C}|$ denotes the total number of semantic categories in the dataset and $\mathcal{C}$ denotes the set of indexes corresponding to the semantic categories.

Our total loss function is composed of four terms, i.e., the cross-entropy loss for the pixel-wise prediction, the guided inter-class loss, the guided intra-class loss, and the classifier weight regularization loss.

The final loss can be expressed as:

$$\mathcal{L}_{final} = \mathcal{L}_{CE} + \underbrace{\beta_1 \mathcal{L}_{inter} + \beta_2 \mathcal{L}_{intra}}_{\mathcal{L}_{GCBL}} + \beta_3 \mathcal{L}_W, \quad (8)$$

where β_1, β_2 , and β_3 are the loss coefficients to balance the contribution of the guided contrastive boundary loss and regularization loss on the main segmentation task loss.

4. Experimental results and analysis

In this section, we compare the performance of semantic segmentation on both image-based and point-based datasets.

4.1. Implementation details

We use STDC [7], SERT [12], SegFormer [8], DeepLabV3 [13], OCRNet [14], and CBL [9] as the baseline models. All experiments are conducted with two NVIDIA V100 GPUS. For the STDC1 and STDC2 models, they are trained with batch size of 16 for 60,000 iterations. For the SERT model, it is trained with batch size of 4 for 40,000 and 80,000 iterations respectively as shown in Table 2. For the Segformer-b0 and Segformer-b5 models, they are trained with batch size of 16 and 4 respectively, for 160,000 iterations. For the DeeplabV3 and OCRNet baselines, they are all trained with batch size of 8 and for 80,000 iterations. We adopt the official code and training setting for CBL baseline. In the experiments, λ is set as 0.1 in both Eqs. (5) and (7). β_1, β_2 , and β_3 are set as 0.1, 0.1, and 0.01 respectively.

It is worth noting that the incremental computational cost of our approach is negligibly compared to standard training time. The primary additional cost arises from the Asymmetric Distance Search operation, which consists mainly of two simple procedures: boundary normal detection and masked feature selection. Our experiments indicate that the increase in training time is only 15%–20%, depending on the specific model.

Table 1

Quantitative comparison of the STDC trained with the official loss (CE + OHEM + binary cross entropy + dice loss) and our CE + GCBL loss. The results are reported on both the validation and test sets of Cityscapes.

Methods	Resolution	Backbone	mIoU (%)		FPS
			val	test	
BiSeNetV1 [34]	768 × 1536	Xception39	69.0	68.4	105.8
BiSeNetV1 [34]	768 × 1536	ResNet18	74.8	74.7	65.5
DF1-Seg-d8 [35]	1024 × 2048	DF1	72.4	71.4	136.9
DF1-Seg [35]	1024 × 2048	DF1	74.1	73.0	106.4
DF2-Seg1 [35]	1024 × 2048	DF2	75.9	74.8	67.2
DF2-Seg2 [35]	1024 × 2048	DF2	76.9	75.3	56.3
BiSeNetV2 [36]	512 × 1024	No	73.4	72.6	156
BiSeNetV2-L [36]	512 × 1024	No	75.8	75.3	47.3
STDC1-Seg50 [7]	512 × 1024	STDC1	72.2	71.9	250.4
STDC2-Seg50 [7]	512 × 1024	STDC2	74.2	73.4	188.6
STDC1-Seg75 [7]	768 × 1536	STDC1	74.5	75.3	126.7
STDC2-Seg75 [7]	768 × 1536	STDC2	77.0	76.8	97.0
STDC1-Seg50 + GCBL	512 × 1024	STDC1	73.0 (↑0.8)	73.8 (↑1.9)	250.4
STDC2-Seg50 + GCBL	512 × 1024	STDC1	74.6 (↑0.4)	74.4 (↑1.0)	188.6
STDC1-Seg75 + GCBL	768 × 1536	STDC1	76.5 (↑2.0)	75.9 (↑0.6)	126.7
STDC2-Seg75 + GCBL	768 × 1536	STDC2	77.9 (↑0.9)	77.4 (↑0.6)	97.0

Table 2

Quantitative comparison of the SERT trained with CE loss and our CE + GCBL loss. The results are reported on validation sets of Cityscapes with different training schedules (e.g., 40k and 80k). ‘Pre’ denotes the backbone is pretrained on the ImageNet-21K dataset.

Methods	Pre	Backbone	#Params	40k	80k
SETR-Base	21 K	T-Base	87.7	75.5	76.3
SETR-Base+GCBL	21 K	T-Base	87.7	76.0 (↑0.5)	76.7 (↑0.4)

4.2. Results on RGB images

4.2.1. Dataset

Cityscapes [15] is a semantic scene parsing dataset comprised of a large, diverse set of stereo videos. It contains 5000 images with high-quality annotations, which were recorded in street scenes from 50 different cities and taken from a car perspective. The numbers of training, validation, and test sets are 2975, 500, and 1525, respectively. Typically, 19 classes are used for semantic segmentation tasks.

ADE20K [16] is a challenging benchmark for semantic segmentation task. It consists of around 20,000 and 2000 images with detailed annotation for training and validation, respectively, including various types of objects and scenes. The dataset includes 150 fine-grained semantic classes.

4.2.2. Competitors and results

STDC [7] is a novel and efficient structure for semantic segmentation task. Based on the carefully designed modules, it achieved a state-of-the-art speed-accuracy trade-off in real-time semantic segmentation.

In **Table 1**, we compare the performance of real-time methods, including BiSeNetV1 [34], DF1-Seg-d8 [35], and BiSeNetV2 [36], using the STDC1 and STDC2 backbones. Our approach consistently improves performance on both validation and test sets. Specifically, for STDC1-Seg50, we achieve improvements of 0.8% and 1.9% on the validation and test sets, respectively. For STDC1-Seg75, we achieve improvements of 2.0% and 0.6%. With STDC2-Seg50, we surpass the baseline by 0.4% on the validation set and 1.0% on the test set. Similarly, for STDC2-Seg75, we achieve improvements of 0.9% and 0.6% on the validation and test sets, respectively.

SERT [12] treated semantic segmentation as a sequence-to-sequence prediction task instead of a conventional encoder-decoder-based FCN architecture. It uses a pure transformer (i.e., without convolution and resolution reduction) to encode an image as a sequence of patches. Since the global context is modeled in every layer of the encoder

Table 3

Quantitative comparison of SegFormer trained with CE loss and our CE + GCBL loss. The results are reported on both the validation and test sets of Cityscapes.

Methods	Encoder	Extra data	val	test
SegFormer-b0	MIT_B0	IM-1K	76.5	–
SegFormer-b5	MIT_B5	IM-1K	82.3	82.2
SegFormer-b0 + GCBL	MIT-B0	IM-1K	77.9 (↑1.4)	–
SegFormer-b5 + GCBL	MIT-B5	IM-1K	82.9 (↑0.6)	82.5 (↑0.3)

Table 4

Quantitative comparison of different competitors trained with and without our GCBL loss. The results are reported on the validation set of ADE20K.

Competitors	STDC1	STDC2	SETR-base	SegFormer-b0	SegFormer-b5
Baseline	20.5 ^a	27.0 ^a	48.1	37.9	49.1
Baseline + GCBL	21.1	27.9	48.9	38.8	49.6

^a Indicates the performance is reproduced using the MMsegmentation codebase [37].

transformer, the model can achieve competitive results with a simple decoder.

In **Table 2**, we report and compare the performance of SERT trained with and without the proposed GCBL loss used on the validation set of Cityscapes. It is worth noting that the model trained with our loss function enhances the mIoU by 0.5% and 0.4%, respectively.

SegFormer [8] is an efficient and powerful semantic segmentation framework that contains a position-encoding-free, hierarchical transformer encoder and a lightweight All-MLP decoder. Multiscale features are output from the hierarchically structured transformer encoder.

In **Table 3**, we report the performance improvement of the state-of-the-art (SOTA) method for semantic segmentation using the SegFormer backbone. The performance improvements are consistent across models, with a particularly significant improvement of 1.4% observed for the simpler model using the MIT-B0 encoder on the validation set.

To further demonstrate the effectiveness of our method, we present the performance improvement of various models on the more challenging ADE20K dataset in **Table 4**. The results indicate that our method effectively enhances the performance of different baseline models even on this more challenging dataset. Specifically, for the STDC1 and STDC2 models, performance improves by 0.6% and 0.9%, respectively, after incorporating our method. The SETR-base model surpasses its baseline by 0.8% with our GCBL loss. Additionally, the SegFormer-b0 and SegFormer-b5 models show improvements of 0.9% and 0.5%, respectively.

To verify the effectiveness of our method across different semantic classes, **Table 5** presents the class-wise performance improvement of

Table 5

Quantitative comparison of class-wise IoU results on the DeepLabV3 and OCRNet models trained with Segfix, IABL, and our GCBL, respectively. The results are reported on the validation set of Cityscapes. Bold indicates the best results, while an underline denotes that the performance trained with our GCBL is greater than or equal to the baselines.

Method	Road	Side-walk	Building	Wall	Fence	Pole	Traffic-light	Traffic-sign	Vegetation	Terrian	Sky	Person	Rider	Car	Truck	Bus	Train	Motorcycle	Bicycle	Mean
DLabV3	98.4	86.5	93.1	63.9	62.6	66.1	72.2	80.0	92.8	66.3	95.0	83.3	65.5	95.3	74.5	89.0	80.0	67.4	78.4	79.5
+Segfix	98.5	87.1	93.5	64.6	63.1	69.0	74.9	82.4	93.2	66.7	95.3	84.9	66.9	95.8	75.0	89.6	80.7	68.4	79.7	80.5
+IABL	98.1	84.9	93.1	59.9	63.1	68.3	75.4	82.3	92.7	64.6	95.0	84.6	69.4	95.6	79.9	87.8	83.3	70.6	80.2	80.5
+GCBL	<u>98.4</u>	<u>86.3</u>	<u>93.2</u>	58.1	<u>62.6</u>	<u>68.2</u>	<u>73.9</u>	<u>81.8</u>	<u>92.8</u>	64.1	<u>95.0</u>	<u>84.3</u>	<u>66.7</u>	<u>95.9</u>	83.4	91.5	86.2	71.3	<u>80.1</u>	80.7
OCR	98.4	86.8	93.3	62.2	66.0	70.0	73.9	82.0	93.0	67.2	95.0	84.3	66.3	95.6	82.9	91.7	85.0	67.7	79.2	81.1
+Segfix	98.5	87.3	93.5	62.6	66.4	71.4	75.7	83.3	93.3	67.6	95.3	85.2	67.2	96.0	83.3	92.2	85.5	68.6	80.0	81.7
+IABL	98.3	86.7	93.6	63.9	64.9	70.2	76.9	83.2	93.2	69.0	95.3	85.6	70.7	96.0	80.0	93.3	86.6	69.3	80.8	82.0
+GCBL	98.5	87.4	93.8	<u>62.7</u>	65.9	<u>72.2</u>	<u>75.6</u>	<u>82.9</u>	<u>93.1</u>	65.5	<u>95.5</u>	<u>84.7</u>	<u>67.5</u>	<u>96.1</u>	87.6	<u>92.1</u>	84.9	<u>70.1</u>	<u>80.5</u>	81.9

Table 6

Experiments for statistical robustness of our method on the Cityscapes val set. we have trained the STDC2 model independently for five times and then calculated the mean and variance of the performance results.

Methods	Run 1	Run 2	Run 3	Run 4	Run 5	mean	std
STDC2-Seg50	74.7	74.7	74.5	74.6	74.5	74.6	0.08
STDC2-Seg75	78.0	77.7	78.0	77.8	77.8	77.9	0.12

models trained with our proposed GCBL. We also compare our approach with other boundary-aware methods, such as SegFix [18] and IABL [22], using DeepLabV3 and OCRNet baselines. On the DeepLabV3 baseline, our method outperforms both SegFix and IABL in terms of mean IoU, achieving a 1.2% improvement over the baseline. While our method falls slightly behind IABL on the OCRNet baseline, it still surpasses the baseline by 0.8%. Examining the class-wise IoU results, we observe that our GCBL method outperforms the baseline for the majority of the classes.

To assess the statistical robustness of our proposed method, we conducted five independent experiments on the STDC2 model. The results, presented in Table 6, demonstrate minimal variation in the final results across different runs. This confirms the robustness of our method against randomness in training.

4.3. Results on 3D point clouds

Additionally, we report the performance of our method on the S3DIS dataset [17], comparing it with the state-of-the-art contrastive boundary learning method [9]. For simplicity, in this experiment, the contrastive learning pairs are selected using the same approach as in the CBL method.

4.3.1. Dataset

S3DIS dataset [17] contains a set of challenging indoor point clouds for semantic segmentation. There are 272 rooms in 6 indoor areas covered by RGB point clouds in 3D. For each point, one of the 13 semantic categories is assigned, e.g., ceiling, floor, and clutter.

4.3.2. Competitors and results

CBL [9] was proposed as a novel and effective contrastive boundary learning framework for point cloud segmentation, which encourages the learned representations to be more similar to their neighbor points from the same class and more distinguished from other neighbor points from different classes. By explicitly optimizing the features on boundaries and improving the model performance on boundaries, the method achieves leading performance and consistent performance improvement across various baselines and datasets.

As shown in Table 7, we compare the performance of the methods including PointCNN [38], SPGraph [39], HPEIN [40], MinkowskiNet [41], KPConv [42], JSENet [43], CGA-Net [44], RandLA-Net [45], CloserLook3D [46] and ConvNet [9]. We observe that the model trained with our GCBL loss outperforms the CBL baseline [9]. Specifically, our GCBL loss yields a 70.0 mIoU, surpassing the baseline

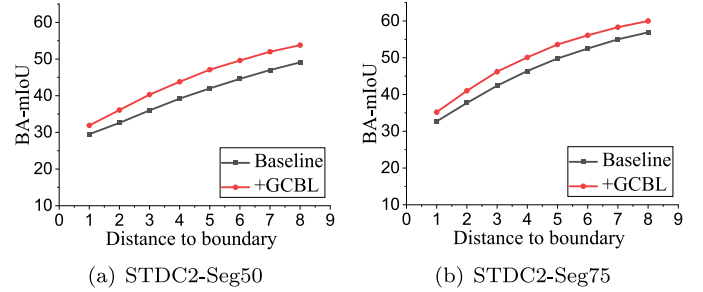


Fig. 6. Illustration of performance changes under different hyperparameters d under the BA-mIoU metric. It can be seen that our method obtains significant performance improvement compared with the baseline model under different distances.

ConvNet by 2.6% and the ConvNet trained with CBL by 0.6%. While our method achieves a 0.1% improvement over the baseline ConvNet in terms of overall accuracy (OA), it falls slightly behind the CBL-trained model by 0.4%. However, for mean average class accuracy (mACC), our method demonstrates a substantial improvement of 3.6% compared to the baseline and 1.2% compared to the ConvNet trained with CBL.

4.4. Compare with other loss functions

To demonstrate the effectiveness of our GCBL loss, we compare its performance with two other loss functions commonly used in segmentation tasks and specifically designed for boundary areas: boundary loss [47] and Lovasz loss [48]. For a fair comparison, we adopt a common combination loss form: $\mathcal{L}_{final} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{other}$, where $\mathcal{L}_{CE}$ is the cross-entropy loss, and $\mathcal{L}_{other}$ represents either the boundary or Lovasz loss. We conduct experiments with λ values of 1.0, 0.1, and 0.01, reporting the best performance for each loss function. * indicates we use the boundary loss proposed in the original STDC paper [7]. The results, presented in Table 8, clearly demonstrate that our GCBL loss consistently outperforms both the boundary and Lovasz loss functions across all evaluated metrics.

To further highlight the advantages of our guided contrastive learning approach, we conducted experiments comparing the performance gain between our method (GCBL) and CBL. As CBL was initially proposed for point cloud semantic segmentation, we adapted the algorithm for image segmentation in this comparison. The results, presented in Table 9, demonstrate consistent performance improvement for models trained with GCBL compared to CBL. Specifically, for the STDC1-Seg50 and STDC2-Seg50 baselines, our approach improves performance by 0.8% and 0.5%, respectively. For the STDC1-Seg75 and STDC2-Seg75 baselines, GCBL surpasses CBL by 1.3% and 0.5% mIoU, respectively.

4.5. Comparison under the BA-mIoU metric

While our primary focus is on improving feature representation learning and segmentation performance at boundaries, the commonly

Table 7

Quantitative results on the S3DIS Area 5 dataset. The mean IoU (mIoU) overall accuracy (OA) and the mean accuracy (mACC) are reported. The methods with * also considers boundaries in their design.

Methods	mIoU	OA	mACC	Ceiling	Floor	Wall	Beam	Column	Window	Door	Table	Chair	Sofa	Bookcase	Board	Clutter
PointCNN	57.3	85.9	63.9	92.3	98.2	79.4	0.0	17.6	22.8	62.1	74.4	80.6	31.7	66.7	62.1	56.7
SPGraph	58.0	86.4	66.5	89.4	96.9	78.1	0.0	42.8	48.9	61.6	84.7	75.4	69.8	52.6	2.1	52.2
HPEIN	61.9	87.2	68.3	91.5	98.2	81.4	0.0	23.3	65.3	40.0	75.5	87.7	58.5	67.8	65.6	49.4
MinkowskiNet	65.4	–	71.7	91.8	98.7	86.2	0.0	34.1	48.9	62.4	81.6	89.8	47.2	74.9	74.4	58.6
KPConv	67.1	–	72.8	92.8	97.3	82.4	0.0	23.9	58.0	69.0	81.5	91.0	75.4	75.3	66.7	58.9
JSENet	67.7	–	–	93.8	97.0	83.0	0.0	23.2	61.3	71.6	89.9	79.8	75.6	72.3	72.7	60.4
CGA-Net	68.6	–	–	94.5	98.3	83.0	0.0	25.3	59.6	71.0	92.2	82.6	76.4	77.7	69.5	61.5
RandLA-Net	62.4	87.2	71.4	91.1	95.6	80.2	0.0	24.7	62.3	47.7	76.2	83.7	60.2	71.1	65.7	53.8
CloserLook3D	66.9	90.0	72.1	94.8	98.4	82.5	0.0	25.5	51.3	70.9	92.1	81.9	76.7	70.1	64.5	61.2
ConvNet	67.4	90.1	72.9	94.1	98.1	83.1	0.0	24.9	53.5	73.0	91.7	82.3	76.5	72.3	66.9	60.8
ConvNet + CBL	69.4	90.6	75.2	93.9	98.4	84.2	0.0	37.0	57.7	71.9	91.7	81.8	77.8	75.6	69.1	62.9
ConvNet + GCBL	70.0	90.2	76.4	95.1	98.0	86.5	0.0	35.3	61.0	72.7	78.9	90.4	83.6	71.6	75.8	60.7

Table 8

Comparison with different loss functions on Cityscapes in terms of mIoU.

Methods	Baseline	Lovasz loss	Boundary loss	Ours
STDC2-Seg50	73.0	73.3	74.2 ^a	74.6
SegFormer-b0	76.5	76.4	76.7	77.9
SERT-Native	75.5	75.1	75.5	76.0

^a Denotes the boundary loss in the STDC method.

Table 9

Quantitative comparison of the STDC trained with CBL and our GCBL loss. The results are reported on both validation sets of Cityscapes.

Methods	Resolution	Backbone	mIoU		
			Baseline	+CBL	+GCBL
STDC1-Seg50	512 × 1024	STDC1	72.2	72.2	73.0
STDC2-Seg50	512 × 1024	STDC2	74.2	74.1	74.6
STDC1-Seg75	768 × 1536	STDC1	74.5	75.2	76.5
STDC2-Seg75	768 × 1536	STDC2	77.0	77.4	77.9

used segmentation metrics assess performance across all image pixels. Since boundary pixels generally constitute a small proportion of the total pixels, their impact on overall segmentation metrics may be limited. Consequently, traditional image segmentation evaluation metrics may not be sufficiently sensitive to reflect improvements specifically in boundary regions.

To more accurately measure the improvement of our method on boundary region segmentation, we propose a boundary-sensitive evaluation metric inspired by the border-IoU proposed in [49]: the Boundary-Aware mean Intersection over Union (BA-mIoU). This metric exclusively considers pixels within a distance d of object boundaries when calculating segmentation performance.

The BA-mIoU metric varies with the hyperparameter d , representing the distance from the boundary. To illustrate performance under different values of d , we present the results as curves in Fig. 6. As expected, the BA-mIoU decreases with diminishing d , reflecting the well-established challenge of accurate boundary segmentation. Conversely, expanding the boundary surrounding area (by increasing d) leads to a corresponding increase in BA-mIoU. With increasing d , the BA-mIoU will gradually converges towards the standard mIoU. Notably, the rate of BA-mIoU increase diminishes as distance increases. This is because the performance difference between the baseline model and our approach primarily manifests near boundaries, and this difference becomes less pronounced as we move away from boundaries.

4.6. Ablation study

We conducted detailed ablation studies on the Cityscapes validation set to examine the effectiveness of each component in our proposed approach using the STDC2-50 model. We performed multiple experiments

Table 10

Ablation study of each component on the final performance on the Cityscapes validation set. ✓ denotes the standard inter-class representation learning.

STDC2-50	Pixel search		G_{inter}	G_{intra}	WR	p-value	mIoU
	SS(4,4)	AS(2,8)					
✓						–	73.1 (±0.17) ^a
✓	✓		✓			0.031	73.7 (±0.21)
✓		✓		✓		0.043	73.5 (±0.09)
✓			✓			0.003	74.1 (±0.14)
✓		✓		✓		0.042	73.4 (±0.05)
✓		✓		✓	✓	0.006	73.9 (±0.12)
✓		✓	✓	✓		0.001	74.3 (±0.12)
✓		✓	✓	✓	✓	<0.001	74.6 (±0.08)

^a Indicate the performance is reproduced results using the official code.

and reported the mean and variance, along with p-values between different settings compared to the baseline. The results are presented in Table 10, where ✓ indicates the standard contrastive learning approach for inter-class representation learning, which pushes apart learned features from different classes with equal strength.

From Table 10, we observe that training solely with the G_{inter} component equipped with an asymmetric distance search strategy yields a significant performance improvement over the baseline, increasing the mIoU from 73.0% to 74.1%. This improvement also surpasses the model trained with the G_{inter} component and symmetric distance search strategy by 0.4%. This highlights the critical role of predictions at the boundary area in the segmentation task and demonstrates the effectiveness of our guided inter-class representation learning in enhancing feature discriminability at boundaries.

Furthermore, the model trained with the standard G_{inter} component and asymmetric distance search strategy achieves a 73.5% mIoU, 0.6% lower than our guided inter-class representation learning method. This underscores the importance of incorporating pixel prediction information in the contrastive learning process.

While the G_{intra} component alone yields a modest 0.4% improvement, we attribute this to the fact that the primary challenge in segmentation lies in learning discriminative representations at boundaries. Therefore, emphasizing inter-class feature differences is more impactful than focusing on intra-class feature similarities. However, using only intra-class loss can lead to feature collapse, and we address this by adding constraints on the weight distribution, leading to further performance gains.

4.7. Visualization

A visual comparison of prediction results between the baseline and the model trained with our guided contrastive boundary learning method is shown in Fig. 7. Our method demonstrates superior results

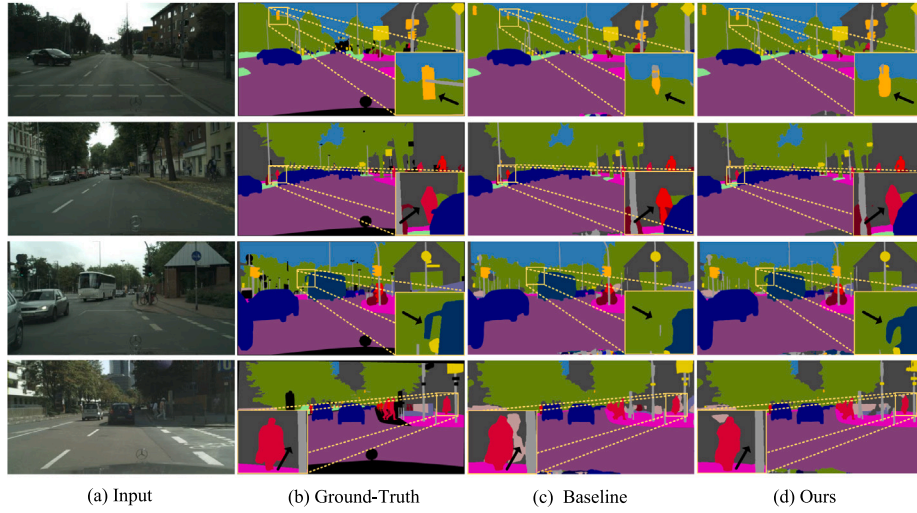


Fig. 7. Visualization comparison of the baseline (STDC2-75) and the model trained with our guided contrastive boundary learning loss. The first column (a) shows the input images, the second column (b) shows the label of the input images, the third column (c) shows the predictions of the baseline, and the last column (d) shows the predictions with our guided contrastive boundary learning loss.

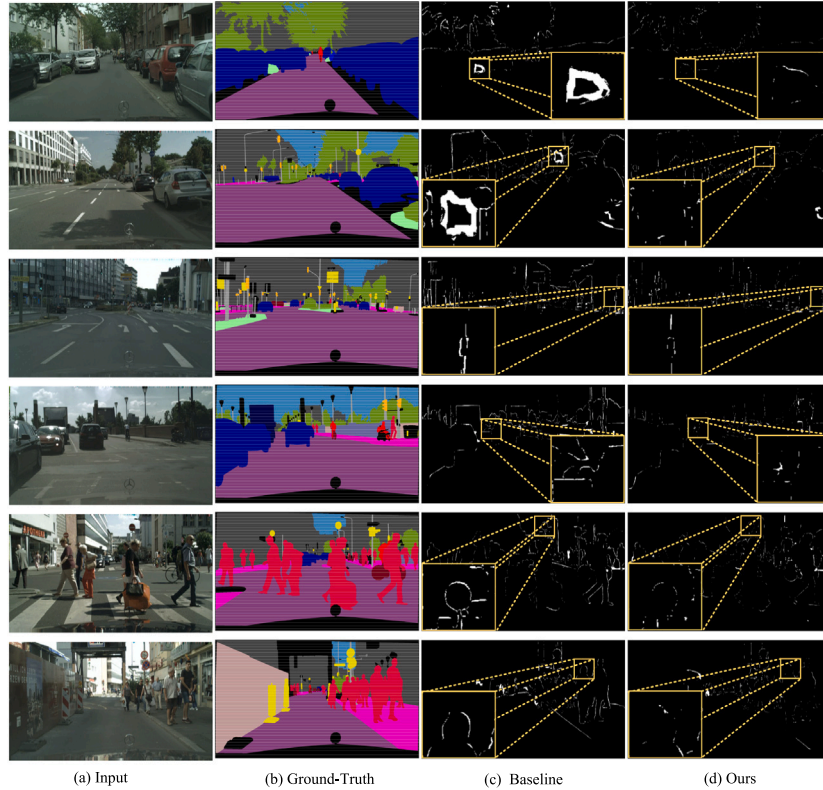


Fig. 8. Visualization comparison segmentation performance at the boundary area of the baseline (STDC2-75) model and the model trained with our guided contrastive boundary learning loss. The first column (a) shows the input images, the second column (b) shows the ground truth of the input images, the third column (c) shows the boundary area where the segmentation result of our model is better than the baseline model, and the last column (d) shows boundary area where our segmentation result of our model is worse than the baseline model. Best viewed with zoom in.

for the detailed prediction of small objects. Segmentation results at boundaries are significantly improved, a crucial aspect for the challenging task of autonomous driving scene understanding. It is important to note that in the second row, although the colors of pedestrians appear all red, there are subtle differences (red and pink). The prediction results obtained after training with our method accurately are consistent with the ground truth color.

To better visualize the differences between the baseline model and our GCBL-trained model in boundary area predictions, we highlight their discrepancies. Since boundary regions are often thin and difficult to directly observe and compare, we preprocess the segmentation results for enhanced visualization.

We define $S_{baseline}$ and S_{GCBL} as binary masks (of size $H \times W$) representing the segmentation errors of the baseline and GCBL models,

respectively. Here, 0 indicates correct predictions, and 1 indicates incorrect predictions. To highlight the differences and ensure a fair comparison, we express the difference between the masks as: $(S_{baseline} \oplus S_{GCBL}) \cap S_{baseline}$, where $\oplus$ denotes the XOR operation, and $\cap$ denotes the intersection operation. The above expression isolates areas where the baseline model made incorrect predictions but our GCBL model made correct predictions, effectively highlighting where our method outperforms the baseline. Similarly, the difference where our method underperforms is given by: $(S_{baseline} \oplus S_{GCBL}) \cap S_{GCBL}$. We set the distance parameter $d = 8$. The results, shown in Fig. 8, demonstrate that our method consistently exhibits fewer segmentation errors in boundary regions compared to the baseline model across various scenarios.

5. Conclusion

In this work, we have introduced Guided Contrastive Boundary Learning (GCBL), a novel and effective approach to enhance semantic segmentation at object boundaries. Our method improves traditional intra-class and inter-class learning by incorporating guidance mechanisms that leverage prediction information and neural collapse phenomena. We also propose an asymmetric distance boundary pixel search algorithm for a more reasonable selection of contrastive pairs. Additionally, a weight regularization loss is introduced to prevent weight collapse during contrastive learning, enhancing model robustness.

Our framework is designed as a plug-in component, easily integrated into existing semantic segmentation methods. Extensive experiments on both image and point cloud segmentation tasks demonstrate substantial and consistent improvements in the mIoU metric.

While our approach shows promising results, it has some limitations. First, the weight of the weight regularization loss is a crucial hyperparameter that requires careful tuning to balance between weight collapse and model convergence. This may pose a challenge in practical applications, necessitating further fine-tuning for different datasets and models. Second, our current pair sampling strategy, focused on boundary regions, lacks explicit consideration for pixel balancing between classes at boundaries.

Future work will explore the application of GCBL to other computer vision tasks, such as video object segmentation and optical flow estimation. Additionally, we will investigate more sophisticated pair sampling strategies to address the issue of class imbalance at boundaries.

CRedit authorship contribution statement

Shoumeng Qiu: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Jie Chen:** Writing – review & editing, Methodology, Conceptualization. **Haiqiang Zhang:** Writing – review & editing, Project administration, Methodology. **Ru Wan:** Funding acquisition. **Xiangyang Xue:** Writing – review & editing, Supervision. **Jian Pu:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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